Darrin Du

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Education

University of Michigan, College of Engineering **Ann Arbor, MI**
Bachelor of Science in Engineering (B.S.E.) in Computer Science May 2028

- GPA: 3.65/4.0 | Honors: Dean's List, Academic Honors, Michigan Achievement Scholarship
- Course Highlights: Data Structures and Algorithms, Operating Systems*, Computer Organization, Computer Vision*

Experience

HAIL Lab - University of Michigan **Ann Arbor, MI**
Undergraduate Research Assistant, Software | Python, OpenCV, ROSBAGs, Git/Github May 2026 - Sept. 2026

- **Sole undergraduate** on a graduate team building VIGIL, a **$26.4M ARPA-H** rural mobile clinic, expanding the real-time Python/OpenCV guidance display with **MS/PhD** mentors, improving clinician accuracy across 3 procedures.
- Maintained continuous eye-protection masking in **99% of tracked frames** during procedures by developing a shoulder-anchored facial tracking system with **segmentation algorithms** and **fallback redundancy**.
- Partnered with hardware and clinical stakeholders to translate ambiguous requirements into concrete technical specs; rapidly prototyped **10+ interactive HTML/web** concepts using **AI-assisted generation workflows**.
- Reduced codebase size and cut runtime by auditing and **re-architecting legacy code** to remove unused components.

NASA SUITS - Collaborative Lab for Advancing Work in Space (CLAWS) **Ann Arbor, MI**
AR Director | Unity, C#, Mixed Reality Toolkit (MRTK) Mar. 2026 - Present

- Co-directed a **12-person engineering subteam** to national selection in the NASA SUITS Challenge **(1 of 10 teams selected)** by overseeing the architecture of an AR astronaut interface and backend.
- Established **Git branching standards** and mandatory pull-request review across the 12-person subteam, cutting cross-team feature integration time from 14 to 5 days.
- Managed a **2-month onboarding protocol** to teach 4 new members HoloLens 2 deployment and Unity best practices.

Projects

User-Level Thread Library
Software Engineer | C++20, Linux, ucontext Sep. 2026 - Oct. 2026

- Co-developed a preemptive user-level **thread library** implementing threads, mutexes, and condition variables with FIFO scheduling, interrupt-safe critical sections, and **multiprocessor support.**
- Managed **thread lifetimes** and stack deallocation using **RAII and smart pointers**; detected API misuse via exceptions and enforced internal invariants with assertions.
- Wrote a **20-case test suite** used for mutation testing against buggy implementations, exposing **80% of injected bugs**.

Clerse
Frontend Developer | Next.js 15, React Flow, TypeScript, Liveblocks, Tailwind CSS Jan. 2026 - Feb. 2026

- Co-developed a **1st-place** spatial AI canvas platform for a 24-hour Claude hackathon, utilizing **Claude and parallel subagents**.
- Built a **"branching" UI canvas** supporting unlimited parallel conversation branches, letting users fork message and AI context threads via right-click and visually track context through animated river edges for **collaborative AI** use.

GEMINI (CLAWS Sub-System / NASA SUITS)
Mixed Reality Developer | Unity, C#, Mixed Reality Toolkit (MRTK), GitHub Nov. 2025 - May 2026

- Chosen over **10 teams** to demo for **former NASA astronaut Anne McClain**, guiding her through 5 lunar mission protocols with an end-to-end AR interface in Unity/MRTK.
- Engineered dynamic UI/UX screen workflows tailored for head-mounted displays (HMDs), **lowering cognitive load** and ensuring stable spatial anchoring under simulated extravehicular activity (EVA) constraints.
- **Bridged the gap between AI and UX** sub-teams, successfully embedding a localized AI agent into the spatial environment of the HoloLens to automate 5 task dispatch and stream **real-time** telemetry.

Skills

- **Programming Languages:** C++, C#, Python, C, MATLAB, HTML/CSS, TypeScript, R, SQL
- **Tools and Technology:** Unity, OpenCV, YOLO, NumPy, PyTorch, Next.js, React Flow, Version Control
- **Soft Skills:** Cross-Functional Collaboration, Communication & Documentation, Decision-making, Debugging